Mathematics Preliminaries

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Mathematics Preliminaries

- You should review the following topics from the Discrete Structures for CS (Discrete Mathematics):
  - Functions
    - Predicates
  - Sets
    - Notations
    - Operations
  - Sequences
  - Proofs
    - Construction
    - Contradiction

Sets

- **Set** is a group of objects represented as a unit.
  - $\{a, b, c\}$
  - $\{1, 2, 3, \dots\}$
  - $\{\}$ or $\emptyset$ (the empty set)
- Characteristic of Sets
  - Unordered
    - $\{a, b, c\} = \{b, c, a\}$
  - No duplicate
    - $\{1, 2, 3, 2\} = \{1, 2, 3\}$
  - Can be finite, infinite, or empty (finite)
- Use $\in$ or $\notin$ to indicate membership status
  - $1 \in \{1, 2, 3\}$
  - $c \notin \{a, b\}$

Functions

- **Function**: $f : D \to R$
  - D is a set (domain)
  - R is a set (range)
  - f associates each element $d \in D$ with some element $r \in R$
    - Denoted by $f(d) = r$.
- **Example**: $f(x) = (x + 1) \mod 3$
  - $D : \{1, 2, 3, 4\}$
  - $R : \{0, 1, 2\}$
  - Associations are as follows:
    - $f(1) = 2$
    - $f(2) = 0$
    - $f(3) = 1$
    - $f(4) = 2$

Predicates

- A **predicate** is a function where its range is $\{true, false\}$
- Example: $f : \mathbb{N} \rightarrow \{true, false\}$

$$f(x) = x \text{ is an even number}$$

  - $f(3) = false$
  - $f(14) = true$

Predicates and Sets

- A set can be defined using a predicate $\mathcal{P}$

$$B = \{x \mid \mathcal{P}(x)\}$$

B is a set that contains all elements x such that $\mathcal{P}(x)$ is true

$$\mathcal{P}(a) = true \text{ if and only if } a \in B$$

- Example:

$$B = \{x \mid x \text{ is an even number}\}$$

  - $B = \{2, 4, 6, 8, \dots\}$

- Example:

$$C = \{(x, y) \mid x, y \in \mathbb{N} \text{ and } x + 1 = y\}$$

  - $C = \{(1, 2), (2, 3), (3, 4), \dots\}$

Predicates and Sets

- Given a set $B = \{x \mid \mathcal{P}(x)\}$, recall that

$$\mathcal{P}(a) = true \text{ if and only if } a \in B$$

  - If $y \in B$, we can conclude that $\mathcal{P}(y)$ is true
  - Similarly, if $\mathcal{P}(y)$ is true, we can conclude that $y \in B$
  - If $z \notin B$, we can conclude that $\mathcal{P}(z)$ is false
  - Similarly, if $\mathcal{P}(z)$ is false, we can conclude that $z \notin B$
- We will encounter these a lot in our proofs

Set Operations

- Make sure to review the following set operations:
  - **Union**: $\cup$
    - $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
  - **Intersection**: $\cap$
    - $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
  - **Complement**:
    - $\overline{A} = \{x \mid x \notin A\}$
  - **Set Different**: $-$ or $\setminus$
    - $A - B = \{x \mid x \in A \text{ and } x \notin B\}$
  - **Cartesian Product**: $\times$
    - $A \times B = \{(x, y) \mid x \in A \text{ and } y \in B\}$
    - Example:

$$\{1, 2\} \times \{a, b\} = \{(1, a), (1, b), (2, a), (2, b)\}$$

  - If A is a set $|A|$ denotes the number of elements in the set A

Functions

- With the notion of the **Cartesian product**, we can define functions that take more than one parameters
- Example: $f(x, y) = \max(x, y)$

$$f : A \times A \to A \text{ where } A = \{1, 2, 3\}$$

  - The domain is $A \times A$
  - The range is A
  - Associations:

$$\begin{aligned} f(1,1) &= 1 \\ f(1,2) &= 2 \\ f(1,3) &= 3 \\ f(2,1) &= 2 \\ f(2,2) &= 2 \\ f(2,3) &= 3 \\ f(3,1) &= 3 \\ f(3,2) &= 3 \\ f(3,3) &= 3 \end{aligned}$$

Sometimes we use table to represent associations

f	1	2	3
1	1	2	3
2	2	2	3
3	3	3	3

Big Union

- Suppose we have sets $A_0, A_1, \ldots A_n$, we can represent the following set

$$A_0 \cup A_1 \cup \cdots \cup A_n$$

using the **Big Union** notation as follows:

$$\bigcup_{i=1}^{n} A_i$$

Note that the condition is for every i where $0 \leq i \leq n$

- Sometimes the condition can be for every element in a set
- Example

$$\bigcup_{r \in \{1,3\}} \delta(r, a) = \delta(1, a) \cup \delta(3, a)$$

Sequences

- A **Sequence** is a list of objects in some order where duplicate is allowed:

$$(a, b, c) \neq (b, c, a)$$

$$(a, b, c, b) \neq (a, b, c)$$

- Finite Sequences are generally called **tuples**
  - A sequence of k elements is a **k-tuple**
  - A 2-tuple usually called an **ordered pair**
- **Note**: $(a, b) = (x, y)$ if an only if $a = x$ and $b = y$.

Alphabets and Strings

- An **alphabet** is a **finite** set of symbols called Σ (sigma)
  - $\Sigma_A = \{\mathrm{a}, \mathrm{b}, \mathrm{c}, \ldots, \mathrm{z}\}$
  - $\Sigma_B = \{0, 1\}$
- A **string** over Σ is a **finite** sequence of symbols from Σ
- The **length** of a string s is the number of symbols it contains, denoted by $|s|$
- Examples:
  - Let $s_1 = (\mathrm{h}, \mathrm{e}, \mathrm{l}, \mathrm{l}, \mathrm{o})$
    - s_1 is a string over Σ_A
    - $|s_1| = 5$
    - For simplicity, $s_1 = \mathrm{hello}$
  - Let $s_2 = (1, 0, 1, 1, 0, 1)$
    - s_2 is a string over Σ_B
    - $|s_2|$ is 6
    - For simplicity, $s_2 = 101101$

Alphabets and Strings

- A string can be empty
  - ε is used to represent the empty string.
- **Concatenation**:
  - Let s and t be strings over Σ, st is the string obtained by appending string t to the end of string s.
  - Example
    - Suppose $s = 101$ and $t = 10$, $st = 10110$
- A string s is a **substring** of a string t if there exists strings u and v such that $t = usv$
  - Note that u and v can be ε

Languages

- A **language** is a set of strings over Σ
  - Can be finite, infinite, or empty ($\emptyset$)
- Example:
  - Let $\Sigma = \{a, b\}$
  - An example of a language B can be

$$B = \{a, b, aa, ab, ba, bb, aaa, aab\}$$

- We generally use **shortlex order** to list a set of strings (shorter strings precede longer string)
- The **lexicographic ordering** which is dictionary ordering may give you a false information
  - Consider listing the set of all strings over $\{a, b\}$:
    - **Sortlex Order**: $\{\varepsilon, a, b, aa, ab, ba, bb, aaa, aab, \dots\}$
    - **Lexicographc Order**: $\{\varepsilon, a, aa, aaa, aaaa, aaaaa, \dots\}$

Boolean Logic

- A boolean value can be either True (1) or False (0)
- Boolean operators

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$	$p \oplus q$
0	0	1	0	0	1	1	0
0	1	1	0	1	1	0	1
1	0	0	0	1	0	0	1
1	1	0	1	1	1	1	0

  - **DeMorgan**: $\neg(p \wedge q) \Leftrightarrow \neg p \vee \neg q$
  - **DeMorgan**: $\neg(p \vee q) \Leftrightarrow \neg p \wedge \neg q$
  - **Implication**: $p \rightarrow q \Leftrightarrow \neg p \vee q$
  - **Equality**: $p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$
  - **Exclusive or**: $p \oplus q \Leftrightarrow \neg(p \leftrightarrow q)$
  - **Distributive**: $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$
  - **Distributive**: $(p \vee q) \wedge r \Leftrightarrow (p \wedge r) \vee (q \wedge r)$
  - **Distributive**: $p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r)$
  - **Distributive**: $(p \wedge q) \vee r \Leftrightarrow (p \vee r) \wedge (q \vee r)$

Proofs

- A typical step in a **proof** is to derive a statement from initial assumptions and hypotheses, or from statements that have been derived previously, or from other generally accepted facts, using principles of logical reasoning
- We will focus on these two types of Proofs
  - Prove by Construction
  - Prove by Contradiction

Proof by Construction

Prove

For every two integers a and b, if a and b are odd, then $a \times b$ is odd.

- The above statement is in the form of $P \rightarrow Q$ where
  - $P = a$ and b are odd
  - $Q = a \times b$ is odd
- Recall the truth table of $P \rightarrow Q$

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

- The only chance that $P \rightarrow Q$ will be false is when P is true and Q is false
- So, we need to show that if P is true, Q is also true (cannot be false)

Proof by Construction

Prove

For every two integers a and b, if a and b are odd, then $a \times b$ is odd.

- Assume that
  - a and b are integers that are ood.
- Need to Derive:
  - $a \times b$ is odd.
- Fact:
  - If x is an integer that is odd, there exists an integer k such that $x = 2k + 1$.
- Since a and b are odd integers, we can derive:
  - $a = 2i + 1$ and $b = 2j + 1$ for some integers i and j

Proof by Construction

- Let $a = 2i + 1$ and $b = 2j + 1$, we have

$$\begin{aligned} a \times b &= (2i + 1) \times (2j + 1) \\ &= 4ij + 2i + 2j + 1 \\ &= 2(2ij + i + j) + 1 \end{aligned}$$

- Thus, $a \times b$ is an odd integer since there exists an integer $k = 2ij + i + j$ such that $a \times b = 2k + 1$.
- **Note**: By showing a couple of test cases is not a proof
  - In the above proof, we assume that a and b can be any odd integers

Proof by Contradiction

- Consider the statement $\neg P \rightarrow$ False
  - Suppose we can show that $\neg P \rightarrow$ False is true
    - Recall that $T \rightarrow F = F$ and $F \rightarrow F = T$
    - Since $\neg P \rightarrow F$ is true, $\neg P$ must be false
    - In other words, P is true.
- Thus, to show that the statement P is True, we need to show that the statement $\neg P \rightarrow$ False is true.
  - From previous example, we need to assume that $\neg P$ is true (or P is false) and try to derive a **false** statement
    - A false statement is a statement that **contradict** with an initial assumption, a hypotheses, a statement that have been derived previously, or another generally accepted fact.

Proof by Contradiction

Prove

$\sqrt{2}$ is an irrational number.

- Fact: An irrational number is any real number that cannot be expressed as a ratio of integers.
  - In other words, if $\sqrt{2}$ is an irrational number, there are no two integers m and n such that $\frac{m}{n} = \sqrt{2}$.
- Let P be the statement "$\sqrt{2}$ is an irrational number"
  - Thus, $\neg P$ is the statement "$\sqrt{2}$ is a rational number"
  - In other words, there exists two integers m and n such that $\frac{m}{n} = \sqrt{2}$.
- For any ration $\frac{m}{n}$ we can obtain $\frac{m}{n} = \frac{p}{q}$ for some positive integers p and q with no common factors greater than 1

Proof by Contradiction

- Let $\frac{p}{q} = \sqrt{2}$ for some positive integers p and q with no common factor greater than 1
- Let's try to derive some statements

$$\frac{p}{q} = \sqrt{2}$$
$$\left(\frac{p}{q}\right)^2 = (\sqrt{2})^2$$
$$\frac{p^2}{q^2} = 2$$
$$p^2 = 2q^2$$

- Since $p^2 = 2q^2$, p^2 is an even number.

Proof by Contradiction

- Since $p^2 = 2q^2$, p^2 is an even number.
- Since p^2 is an even number, p is an even number
- Since p is an even number, $p = 2r$ for some integer r
- Keep deriving:

$$\begin{aligned} p^2 &= 2q^2 \\ (2r)^2 &= 2q^2 \\ 4r^2 &= 2q^2 \\ 2r^2 &= q^2 \end{aligned}$$

- Since $q^2 = 2r^2$, q^2 is an even number
- Since q^2 is an even number, q is an even number

Proof by Contradiction

- Earlier we stated that $\frac{p}{q} = \sqrt{2}$ for some positive integers p and q with no common factor greater than 1
- But we show that if $\sqrt{2}$ is a rational number both p and q are even
- Therefore p and q have a common factor greater than 1 which is **False** according to our assumption
- Therefore, P is true
- In other words, $\sqrt{2}$ is an irrational number.

Proofs in CS1502

- There will be a lot of proofs in this class (Formal Methods)
  - Hardly any mathematical facts or theorems will be used
  - The type of proof to use is predictable
    - To do this, use prove by construction
    - To do that, use prove by contradiction